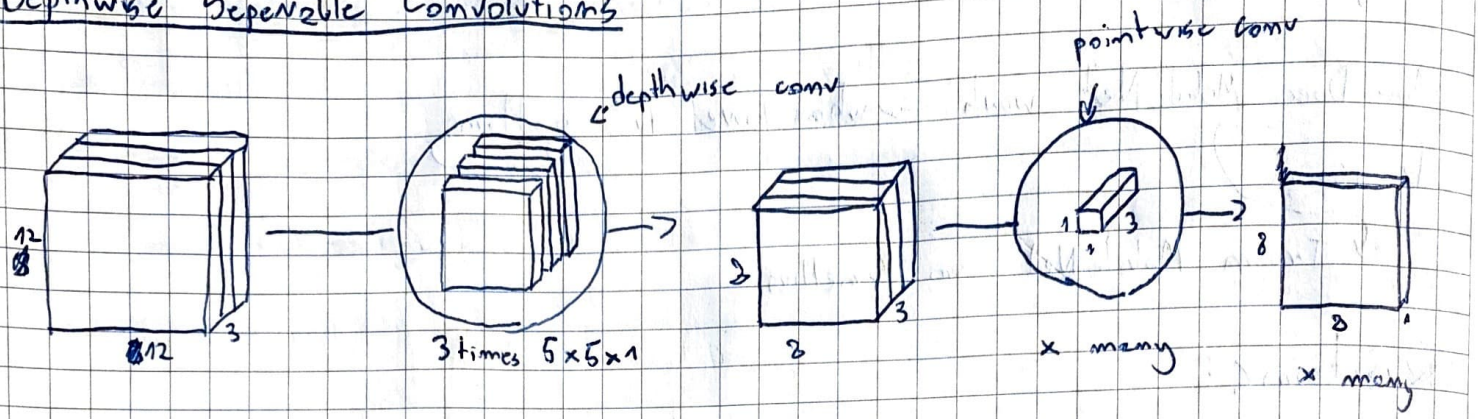
MobileNetV2 Paper:

Depthwise Seperable Convolutions



Way more efficient than 5x5xX, and work almost as well

↳ almost factor of channel^2 - for RGB - 8-9

"Manifold of Interest" - the space of possible activation tensors from imp real input

↳ geometric representation of data e.g. possible images

output of NN after activation function

↳ is smaller than the actual space - people believe can be encoded in smaller space - MobileNetV1 uses dimensionality reduction - computation/Accuracy trade-off

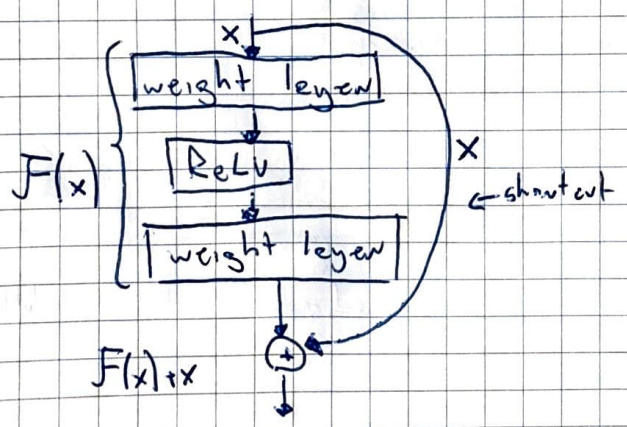
↳ By this assumption they introduce a linear bottle neck

Inverted Residuals

Background:

Vanishing/Exploding gradient - after increasing layers too much the gradient either becomes 0 or too large resulting in worse train and test accuracy

↳ ResNet - Microsoft 2015 - introduced Residual Blocks with skip connections



These do wide → narrow → wide (skip)

Inverted residual blocks - narrow → wide → narrow